



Using Social Media to Disseminate Effective Pain Treatments for Newborns During Needle-Related Painful Procedures in China

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ABSTRACT

Social media has become a powerful approach to disseminating evidence to knowledge users. The BSweet2Babies video was developed in multiple languages showing the effectiveness of sweet solutions, skin-to-skin care, and breastfeeding during newborn painful procedures. This study aimed to disseminate the BSweet2Babies video in Chinese through social media platform of WeChat in China; evaluate the reach, acceptability, and recommendation of the video; and assess viewers' previous knowledge and experience of using the 3 strategies and intention to use these strategies in the future. Multiple dissemination strategies were used to maximize views for a 6-month dissemination period. The video received 19 812 views, 4306 "thumbs," and 671 participants completed surveys. Of the survey respondents, 393 were parents. Most respondents did not know these strategies and did not use or help parents use any of them. More healthcare professionals than parents intended to use or advocate for sweet solutions and breastfeeding. More healthcare professionals rated that the 3 strategies were easy to apply in real-life situations, but more parents evaluated that the length of the video was too long. Social media in China can be a promising approach to disseminating evidence on neonatal procedural pain treatments to healthcare professionals and the public.

Key Words: evidence-based practice, neonate, pain management, procedural pain, social media

Nearly all neonates receive blood sampling by venipuncture or heel lance for newborn screening in their first days of life¹ and preterm and sick newborns admitted to intensive care units require repeated needle-related painful procedures for medical treatments or diagnoses.² A systematic review of the epidemiology of painful procedures illustrated that each sick neonate underwent an average of 7.5 to 17.3 painful and distressing procedures per day in the first 14 days of life.² More recent studies have consistently shown that neonates undergo numerous painful procedures during the hospitalization period.³⁻⁶

Poorly treated procedural pain may have significant short-term and long-term adverse physiological and psychological effects, potentially leading to increased risks of disability and suffering later in life. Untreated pain in infants may also have long-term effects on their subsequent pain perception and responses, neurodevelopment, brain development, and programming of stress systems.⁷⁻⁹ A systematic review included 13 studies examining the association between early pain experiences of preterm infants and subsequent developmental outcomes.⁸ The results showed that the number of painful procedures was the strongest predictor of poorer cognitive and motor development and altered temperaments, such as reduced ability to self-regulate and return to baseline functioning, poorer quality of focused attention, and high levels of internalizing behavior in later life.⁸

Evidence shows that there are 3 very effective, simple, and low-cost strategies to minimize pain in newborns during painful procedures. High-quality synthesized evidence shows that giving neonates small amounts of sweet solutions (sucrose or glucose) with or without a pacifier, skin-to-skin care (SSC), or breastfeeding effectively and safely reduces pain in newborn infants during needle-related painful procedures.¹⁰ A systematic review of 168 studies, which included 62 studies in a cumulative meta-analysis, showed that evidence to support analgesic effects of sweet solutions compared with no treatment or placebo has existed since the year 2000.¹¹ Cochrane systematic reviews have consistently shown that SSC reduces pain responses to needle-related painful procedures,^{12,13} and systematic reviews show that breastfeeding reduces procedural pain in infants compared with no treatment or interventions such as topical anesthetics, maternal holding, and small volumes of expressed breast milk.^{14,15}

Despite the strong evidence base of these effective, feasible, and safe pain management strategies, numerous studies show infrequent use of these strategies during needle-related painful procedures in many settings.² This knowledge-to-action gap has existed for many years and has not changed with the advance

in the evidence on pain treatments. For example, a provincial-wide survey in 2015 in Canada ascertained that more than half of newborn units in Ontario never or occasionally used sweet solutions, SSC, or breastfeeding during heel lance or venipuncture.¹⁶ Two recent Canadian studies published in 2019 found that sucrose was not systematically used in newborns despite sucrose being considered a standard pain management intervention.^{3,17} Pain management interventions requiring the presence of parents (eg, SSC, breastfeeding) were never used, and no pain treatments were used during around 40% of painful procedures.^{3,17} This issue was even more worrying outside the high-income Western countries.² For example, one study in China showed no pain treatment used for a cohort of 108 neonates during 10 633 painful procedures.¹⁸

One of the common barriers to knowledge translation in non-English-speaking countries, including China, is a lack of access to information and knowledge about evidence-based practices.¹⁹⁻²¹ Social media has however, become a powerful approach to widely disseminating evidence to knowledge users, including not only healthcare professionals (HCPs) and organizational leaders but also patients and family members.^{22,23} Harrison et al²⁴ developed a video named *Be Sweet to Babies* (abbreviated to *BSweet2Babies*) in multiple languages, including Chinese, to disseminate knowledge about 3 key evidence-based pain management strategies (ie, sweet solutions, SSC, and breastfeeding) for newborns during blood work. The video in English was first posted on YouTube (<https://youtu.be/L43y0H6XEh4>) on July 31, 2014. After a 1-year dissemination period, the video received 10 879 views, 31 likes, and 187 survey responses.²⁴ Most respondents answered that the video was a helpful resource (92%), was the right length (84%), and was easy to understand (92%).²⁴

However, the *BSweet2Babies* video in Chinese had limited dissemination on YouTube with only 530 views, 0 likes, and 11 survey responses from February 3, 2015, to September 15, 2017.²⁵ One of the reasons might be that YouTube is not accessible in China. Therefore, a potential solution is to use the popular social media platform, WeChat, in China to disseminate high-quality evidence to stakeholders.¹⁹ WeChat is one of the most popular social media platforms in China. This multipurpose application was first released by the Chinese multinational company Tencent Holding Limited in 2011.²⁶ WeChat provides numerous services, including individual or broadcast (one-to-many) messaging, chat groups, and posting in the WeChat Moments.²⁷ The function of WeChat Moments is similar to the Twitter-sharing page and for users to share texts, photographs, and videos, as well as comments and likes.²⁷ Using WeChat has become a daily routine for many

Chinese people. In 2019, it had more than 1.1 billion monthly active users who spent 77 minutes on the app and checked the “WeChat Moments” 10 times daily on average.²⁶ Thus, WeChat was used in this study to widely disseminate the Chinese BSweet2Babies video in China as well as to study knowledge and use of the recommended pain management strategies and the potential effectiveness of the video in practice.

OBJECTIVE

The objective of the present study was to disseminate the Chinese BSweet2Babies video through social media in China and evaluate the reach, acceptability, and recommendation of the video. Additional objectives were to assess viewers’ previous knowledge and experience of using the 3 strategies and intention to use these strategies in the future.

METHODS

Dissemination strategies

A post was developed in WeChat, which contained the Chinese BSweet2Babies video and a brief paragraph introducing the video.²⁸ The video in Chinese is 5 minutes 55 seconds in duration, with the following segments: Introduction (0:00-0:32), Breastfeeding (0:33-1:58), SSC (1:59-3:44), sweet solutions (3:45-5:11), and acknowledgments (5:12-5:55).²⁸ A comprehensive evidence dissemination plan was put into place to maximize views during 6 months of being posted on WeChat. First, the WeChat post was shared by the research team in the WeChat Moments. Second, the post was sent via WeChat messages to others for reading or further dissemination. Third, the post was recommended to different chat groups in WeChat who may be interested in these pain treatments, for example, national or local nursing associations, children’s hospitals, and new mothers’ groups. Fourth, the quick response (QR) code for linking the poster was shared on relevant publicly available Web sites in China (eg, Sohu Web page) or blogs (eg, Fudan Center of Evidence-Based Practice).

Data collection

During 6 months of being posted on WeChat, the daily number of views and likes was assessed and documented in a process log. In addition, the WeChat post contained a link to an online survey following the Chinese BSweet2Babies video and the brief introduction. Research Electronic Data Capture (REDCap) was used to create and distribute the online survey.²⁹ The survey contained 11 brief close-ended questions, which

asked the respondent’s role (ie, parent, HCP), the specific role of HCPs, if applicable (eg, physician, nurse, researcher), and how the WeChat post was accessed (eg, WeChat message, WeChat Moments, QR code). These demographic questions were followed by questions regarding previous viewing of the BSweet2Babies video, prior knowledge of procedural pain treatments for newborns (ie, sweet solutions, SSC, breastfeeding), previous use of each treatment, future intention to use each strategy, intention to recommend the video, perception of the length of the video, and perceptions of how helpful or beneficial, easy to understand, and easy to apply in real-life situation the video is. These questions were translated from those in the previous studies, which have been validated and piloted.^{24,30} In addition, these translated survey questions were also piloted with the research team and a pilot sample of 5 Chinese-speaking parents and 5 Chinese-speaking nurses for clarity and content.

Data analysis

The IBM SPSS for Windows Version 25 statistical package (IBM Inc, Chicago, Illinois) was used for statistical analyses. All quantitative data were analyzed using descriptive statistics and reported using frequencies in numbers and percentages. A χ^2 statistical test was performed to assess the different perceptions of parents and HCPs on the 3 pain management strategies and the video. Statistical significance was determined by both the probability of type I error (α), which was set at 0.05, and the effect size of the statistical test, which was determined using the value of ϕ greater than 0.15.^{31,32}

Ethical approval

Nurses and parents recorded in the video have given written consent to have the video posted online. Ethical approval to post the translated video and conduct this study was granted by the affiliated hospital’s research ethics board (protocol #14/108X).

RESULTS

During the 5-month study period from September 15, 2017, to March 14, 2018, the WeChat post received 19 812 views, 4306 likes, and 671 participants completed surveys (3.39% response rate). Of all the survey respondents, 393 participants identified themselves primarily as parents (58.57%). The remaining survey respondents identified themselves as HCPs ($n = 278$, 41.43%), including physicians ($n = 125$, 18.63%), nurses ($n = 118$, 17.59%), researchers ($n = 12$, 1.79%), faculty ($n = 12$, 1.79%), healthcare administrators ($n = 7$, 1.04%), and other healthcare providers ($n = 4$, 0.60%).

Table 1. Previous knowledge, previous experience, and future intent-to-use evidence-based pain management strategies for newborns during needle-related procedures

	HCPs, <i>n</i> (%)	Parents, <i>n</i> (%)	Total, <i>n</i> (%)	χ^2 test
Previous knowledge of the following pain treatments for newborns before				
Sweet solutions	35 (12.59)	10 (2.54)	45 (6.71)	$\chi^2 = 26.26, P < .001, \phi = -0.20$
Skin-to-skin care	82 (29.50)	54 (13.74)	136 (20.27)	$\chi^2 = 25.01, P < .001, \phi = -0.19$
Breastfeeding	89 (32.01)	82 (20.87)	171 (25.48)	$\chi^2 = 10.66, P = .001, \phi = -0.13$
Any of these 3 strategies	142 (51.08)	117 (29.77)	259 (38.60)	$\chi^2 = 31.19, P < .001, \phi = -0.22$
Previous experience of using or helping parents use the following pain treatments for newborns before				
Sweet solutions	15 (5.40)	3 (0.76)	18 (2.68)	$\chi^2 = 13.38, P < .001, \phi = -0.14$
Skin-to-skin care	50 (17.99)	48 (12.21)	98 (14.61)	$\chi^2 = 4.35, P = .037, \phi = -0.08$
Breastfeeding	32 (11.51)	67 (17.05)	99 (14.75)	$\chi^2 = 3.97, P = .046, \phi = 0.08$
Any of these 3 strategies	74 (26.62)	107 (27.23)	181 (26.97)	$\chi^2 = 0.03, P = .861, \phi = 0.01$
Future intent to use or advocate for the following pain treatments for newborns				
Sweet solutions	227 (81.65)	146 (37.15)	373 (55.59)	$\chi^2 = 130.62, P < .001, \phi = -0.44$
Skin-to-skin care	243 (87.41)	369 (93.89)	612 (91.21)	$\chi^2 = 8.53, P = .003, \phi = 0.11$
Breastfeeding	220 (79.14)	203 (51.65)	423 (63.04)	$\chi^2 = 52.78, P < .001, \phi = -0.28$
Any of these 3 strategies	271 (97.48)	388 (98.73)	658 (98.06)	$\chi^2 = 1.44, P = .230, \phi = -0.05$

Abbreviation: HCPs, healthcare professionals.

Most survey respondents ($n = 437$, 65.13%) viewed the WeChat post, including the BSweet2Babies video, through WeChat message, 216 participants (32.19%) were through WeChat Moments, and 18 participants (2.68%) accessed the video and survey through scanning the QR code. Only 14 (2.1%) survey respondents had seen the Be Sweet to Babies video before. They had watched the video on other video Web sites ($n = 10$, 1.49%), other WeChat posts ($n = 2$, 0.30%), or in conferences ($n = 2$, 0.30%).

Table 1 shows the participants' knowledge, previous use, and future intent-to-use evidence-based pain management strategies for newborns during needle-related procedures. More than 60% of respondents ($n = 412$, 61.40%) did not know about sweet solutions, SSC, or breastfeeding for pain management, and more than 70% of respondents ($n = 490$, 73.03%) did not use or help parents use any of these strategies. Sweet solutions were the least known ($n = 45$, 6.71%) and used ($n = 18$, 2.68%) pain management strategy. The percentage of HCPs who knew these pain management strategies is statistically significantly higher than parents, especially for sweet solutions ($\chi^2 = 26.26, P < .001, \phi = -0.20$) and SSC ($\chi^2 = 25.01, P < .001, \phi = -0.19$). However, numbers of HCPs and parents who used or helped parents use these pain management strategies were not statistically different. Almost all participants intended to use or advocate for SSC ($n = 612$, 91.21%), and the percentages of HCPs and parents were equal. More than half of the participants intended to use or advocate for sweet solutions ($n = 373$, 55.59%) or breastfeeding ($n = 423$, 63.04%) to reduce procedural pain for newborns. The percentage of HCPs intending to use or advocate for these 2 pain treatments was statistically significantly

higher than parents (sweet solutions: $\chi^2 = 130.62, P < .001, \phi = -0.44$; breastfeeding: $\chi^2 = 52.78, P < .001, \phi = -0.28$).

Illustrated in Table 2, nearly all respondents would recommend the video and found that the video was easy to understand and helpful or beneficial for them. However, nearly 75% of respondents felt that the video was too long, and just less than 40% felt that the strategies were not easy to apply in real-life situations. In addition, HCPs and parents had statistically significantly different perceptions of these 2 indicators. More HCPs rated that these strategies were easy to apply in real-life situations ($\chi^2 = 185.02, P < .001, \phi = -0.53$), but more parents evaluated that the length of the video was too long ($\chi^2 = 76.54, P < .001, \phi = 0.34$).

DISCUSSION

Over the six-month period of dissemination, the Chinese BSweet2Babies video reached almost 20 000 views and more than 4300 likes, suggesting that WeChat has been able to reach a large number of viewers to disseminate evidence on procedural pain treatments for neonates in China. Other studies in China have also used WeChat in their knowledge dissemination strategies. One study developed and evaluated a WeChat-based evidence resource platform to improve nurses' access to evidence-based practice in China.³³ They disseminated 223 posts over 3 years and received 584 834 views in total, an average number of views per post of 2621. Another study reviewed the current WeChat accounts of the Centers for Disease Control and Prevention for public health education.³⁴ They identified 50 939 posts, which received 18.66 million views

Table 2. Participants' perceptions on the Chinese BSweet2Babies video

	HCPs, <i>n</i> (%)	Parents, <i>n</i> (%)	Total, <i>n</i> (%)	χ^2 test
Would recommend to others	266 (95.68)	382 (97.20)	648 (96.57)	$\chi^2 = 6.60, P = .037, \phi = 0.10$
Easy to apply	258 (92.81)	162 (41.22)	420 (62.59)	$\chi^2 = 185.02, P < .001, \phi = -0.53$
Easy to understand	273 (98.20)	386 (98.22)	659 (98.21)	$\chi^2 = 0.00, P = .99, \phi = 0.00$
Helpful	276 (99.28)	391 (99.49)	667 (99.40)	$\chi^2 = 0.12, P = .727, \phi = 0.01$
Too long	158 (56.83)	341 (86.77)	499 (74.37)	$\chi^2 = 76.54, P < .001, \phi = 0.34$

Abbreviation: HCPs, healthcare professionals.

and more than 306 000 likes over 2 years (ie, the average number of views = 366, the average number of likes = 6). The larger number of views for the Chinese BSweet2Babies video may be due to many factors, including the strategically planned extensive dissemination strategies. This corroborates the difference from diffusion to dissemination, whereas diffusion is a passive and largely unplanned spread of evidence and dissemination is the conscious and planned effort to spread evidence to target audiences or the public at large.³⁵

The WeChat platform has been commonly used to deliver online surveys to target populations due to the advantages of using social media: speed, reach, ease, and low cost.^{36–38} However, the response rate to the online survey was low in this current study, only 3.4% of viewers. Such low response rates to online surveys have been identified in other studies that have used YouTube to disseminate videos. For example, the BSweet2Babies video in English had a survey response rate of only 1.7%),^{24,30} and the “Power of a Parent’s Touch” response rate was 0.13%.³⁹ One of the reasons for the low questionnaire response rates may be that the survey link was able to be accessed at the end of the video, thereby relying on the fact that the video needed to be viewed all the way through. Yet, it has been shown that people have reduced attention span when using social media.⁴⁰ In addition, the studies of the English video found that the videos are not watched all the way through to the end.^{24,30,39}

In this current study, nearly 400 survey respondents (more than half) were parents. This is more than that found in a systematic review of 12 studies disseminating evidence-based pediatric pain-related knowledge to parents, in which most respondents were HCPs.⁴¹ Neonatal care is considered a partnership between the healthcare professions and the family, and breastfeeding and SSC during painful procedures are parent-led pain treatments. Thus, it is important to disseminate evidence regarding painful procedural management to the family and the parents. This study suggests that WeChat can be a promising approach to reaching parents and increasing parents’ awareness of the role they play in re-

ducing their child’s pain in the Chinese community. In addition, the result about the participants’ approaches to the WeChat post in the study showed that disseminating the WeChat post through chat groups, which share similar and related interest, and through WeChat moment was much more efficient than only sending or posting a QR code of the WeChat post.

Although the survey showed that more HCPs knew the evidence concerning pain treatments for newborns during needle-related pain procedures, the awareness rate in both groups was low and the experience of using, or helping parents use these treatments, was significantly limited. The awareness rate in this study population is much lower than the result identified in the study by Harrison et al²⁴ and the study by Campbell-Yeo et al.³⁹ This finding may be due to many reasons, including lack of pediatric procedural pain management in the Chinese HCP education program, lack of related guidelines in the Chinese context, and language obstacles. The language barrier has been reported in other studies in China and other non-English-speaking countries.^{19,42–45} The majority of evidence synthesis on pediatric procedural pain treatments have been published in English^{11,12,14,15}; however, less than 1% of people in China are able to fluently understand English.¹⁹

Most HCPs had high intentions to use or advocate for all the evidence-based procedural pain treatments. Most parents intended to use SSC; however, fewer parents were willing to use sweet solutions and breastfeeding. A previous study has explored HCPs’ perceived barriers and facilitators to effective procedural pain treatments for pediatric patients in the Chinese context.⁴⁶ Nurses reported that most parents in China wish their children to avoid sweet food early in their life.⁴⁶ However, this requires parental education about the analgesic effects of small volumes of sweet solutions in the context of procedural pain management, which is unrelated to sweet food. In terms of breastfeeding, 2 previous studies identified the availability of parents as a barrier, especially when routine blood sampling was performed early in the morning.^{16,46} These findings suggest that future implementation strategies should address these common

barriers to promote the engagement of parents in procedural pain management for neonates.

Overall feedback on the Chinese BSweet2Babies video was positive from HCPs and parents in the study. But more parents considered that these strategies were not easy to apply and the video was too long. Further parent-targeted educational videos need not only to show the effectiveness of these pain treatments but also to introduce the importance of reducing procedural pain in neonates, the mechanism of these strategies, and user-friendly instructions of using these strategies. Such videos could provide parents with a better understanding of procedural pain and the treatments and help facilitate their use of these strategies. Yet, future videos need to be brief (ie, 1 minute to 3 minutes) and have engaging key messages.^{24,30,39}

PRACTICE IMPLICATIONS

Using social media to disseminate evidence on pediatric pain management has the potential to reach large audiences, making this a promising evidence dissemination strategy for the public in the Chinese context. Social media have also been widely utilized to disseminate evidence or research findings to HCPs and the public in other healthcare settings, beyond China.^{47–49} However, the choice of social media platform needs to be considered on the basis of the needs of the target populations. Using an appropriate social media platform will significantly increase the likelihood of communicating and interacting with wider audiences and facilitating evidence use in decision-making processes and practice.

The Chinese BSweet2Babies video shows how sweet solutions, SSC, and breastfeeding can effectively and safely reduce pain in newborns during needle-related painful procedures; however, it is important to include neonatal procedural pain management in Chinese HCP education programs. In addition, new brief parent-targeted educational videos need to be codeveloped with parents to ensure these videos' relevance based on the parents' knowledge and beliefs. These videos can focus on introducing the importance of reducing procedural pain in neonates, the mechanism of using sweet solutions, SSC and breastfeeding for procedural pain treatments, and user-friendly instructions for using these strategies. Further knowledge translation initiatives are needed to change contextual factors and to transfer HCP's high intentions into behaviors—that is, actual use of these strategies in clinical settings.

STRENGTHS AND LIMITATIONS

The large and widespread reach of the Chinese BSweet2Babies video in this study demonstrated the

usefulness of evidence dissemination using social media in China. However, detailed information on the duration of video viewing was unable to be accessed through WeChat. It is, therefore, not possible to assess whether the videos were watched all the way through, how the information had been delivered to the viewers, and how the viewers engaged in the evidence dissemination. Second, the comparisons on HCPs' and parents' perceptions of the video and their previous knowledge and experience of using the 3 strategies and intention to use these strategies in the future provided new knowledge in this field and informed further investigations. However, a limitation to the study is the low response rate to the online survey, based on the number of video views, which may be indicative of a response bias.

CONCLUSION

Disseminating evidence on procedural pain treatments for newborns using the Chinese BSweet2Babies video through the social media platform of WeChat in China resulted in widespread reach to HCPs and parents. Most HCPs and parents were not aware of and had not used the pain reduction strategies. Intention to use these strategies in the future and feedback on the video were positive. However, most parents did not intend to use sweet solutions or breastfeeding for neonatal procedural pain treatments and commented that the strategies were not easy to apply and the video was too long. Further knowledge translation initiatives that increase parents' awareness and change contextual factors of using procedural pain treatments in neonates are warranted.

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